Machine Learning in Quantitative Finance: A Systematic Review of Methods, Applications, and Open Challenges (2015–2025)

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Abstract

We review 227 studies applying machine learning to quantitative finance, published between 2015 and 2025, and organize them into a two-dimensional taxonomy covering six application domains (return prediction, risk management, portfolio optimization, derivatives pricing, market microstructure, and alternative data) and five methodological families (tree-based methods, deep learning, reinforcement learning, NLP, and hybrids). The literature reveals three generational shifts: gradient-boosted trees establishing dominance on tabular financial data (2015–2018), attention mechanisms and transformers displacing recurrent networks (2018–2022), and large language models entering financial applications (2022–2025). Our main findings are that tree-based ensembles remain hard to beat for cross-sectional prediction, that the strongest ML improvements over traditional methods are in derivatives pricing and volatility forecasting rather than return prediction, that reinforcement learning for portfolio management has attracted more papers than evidence, and that only about one in five studies shares code. We provide standardized comparison tables of selected representative studies across all six domains, propose a Reproducibility Disclosure Score (RDS) and apply it to 99 empirical studies in our database (mean 0.78 on a 0–2 scale; only 16% disclose both code and openly accessible data), and release the full annotated 227-study database alongside this paper as supplementary material. We conclude with ten open problems, arguing that the field's most pressing need is not better models but better benchmarks, reproducibility standards, and audit-grade artifacts for regulator-defensible deployment.

Keywords: machine learning, quantitative finance, systematic review, deep learning, return prediction, portfolio optimization, risk management, NLP in finance

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1 Introduction

Ten years ago, applying machine learning to financial markets was a niche pursuit. A handful of papers tested random forests or SVMs on stock returns, typically on a single market and a single horizon, and the results were mixed enough that most finance departments ignored them. That is no longer the case. By 2025, ML has touched every corner of quantitative finance: cross-sectional asset pricing, options hedging, sentiment extraction from earnings calls. The volume of published work has grown to a point where keeping track of it is itself a research problem.

The trouble is that this body of work is scattered. An ML researcher looking for the best model to predict equity returns will find relevant papers at NeurIPS, in the *Review of Financial Studies*, on arXiv's q-fin section, and buried in SSRN working paper series. The conventions are different in each venue. ML papers report out-of-sample R^2 and treat a 0.5% improvement as a win. Finance papers want to see Sharpe ratios net of transaction costs and a comparison against Fama-French factor models. A result that looks impressive by one community's standards may look unremarkable, or even suspicious, by the other's. This fragmentation wastes effort: researchers in one camp routinely rediscover things the other camp established years ago.

1.1 What This Survey Does

This paper attempts to impose some order on the chaos. We systematically review ML applications in quantitative finance from 2015 through 2025, covering 227 studies across six application domains: return prediction, risk management, portfolio optimization, derivatives pricing, market microstructure, and alternative data/NLP. We classify each study along two dimensions, methodology and application domain, producing a taxonomy matrix that makes it easy to see where the literature is dense and where it has gaps.

Three things distinguish this survey from its predecessors:

1. **Standardized comparison tables.** For each domain, we report the dataset, sample period, ML method, benchmark, performance metric, and (the field's blind spot) whether the code and data are publicly available. Most prior surveys describe methods in prose; we tabulate them so readers can compare directly.

2. **Full coverage of the LLM era.** Existing surveys largely predate the 2022–2025 explosion of large language models in finance. We cover BloombergGPT, FinBERT, GPT-based return prediction, and the broader question of whether foundation models will reshape the field.

3. **A reproducibility audit.** We track, for each study, whether code and data are available. The numbers are not encouraging: roughly 21% of papers share code. We

flag this not as an aside but as a first-order problem for the field.

1.2 Prior Surveys and Where They Fall Short

The literature already has several surveys, each useful but each with blind spots. Henrique et al. (2019) cover ML for stock prediction through 2018, but that cutoff misses transformers, LLMs, and the post-pandemic data regime entirely. Ozbayoglu et al. (2020) survey deep learning architectures for finance, though their emphasis is on describing network structures rather than comparing empirical results head-to-head. de Prado (2018) and Dixon et al. (2020) wrote valuable textbooks, but textbooks age, and neither could anticipate the pace of change after 2020.

Among more recent work, Kelly and Xiu (2023) provide the most authoritative treatment: their *Foundations and Trends in Finance* monograph is close to definitive on return prediction and factor models. But their scope is deliberately narrow; they do not cover NLP, derivatives pricing, or microstructure. Nazareth and Reddy (2023) cast a wider net (126 articles, 44 journals) but focus on categorization rather than critical evaluation. They tell you what methods have been applied where, not which ones actually work. Goodell et al. (2022) take a bibliometric approach (348 papers, co-citation analysis) that maps the intellectual landscape but provides no guidance to a practitioner wondering which model to deploy.

Our survey fills the gap between these approaches: broad enough to cover all six domains, deep enough to include comparison tables and reproducibility tracking, and current enough to address the LLM wave.

1.3 Roadmap

Section 2 describes our search strategy and inclusion criteria. Section 3 lays out the two-dimensional taxonomy. Section 4 through Section 9 work through each application domain in turn. Section 10 tackles themes that cut across domains: interpretability, non-stationarity, compute, and the reproducibility crisis. Section 11 identifies ten open problems. Section 12 steps back and offers an overall assessment.

2 Systematic Review Methodology

We follow the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines adapted for computational research (Page et al., 2021). This section details our search strategy, selection criteria, and data extraction protocol.

2.1 Search Strategy

We conducted systematic searches across five databases:

- **Scopus** — comprehensive coverage of finance and CS journals
- **Web of Science** — primary citation database for impact tracking
- **SSRN** — working papers and preprints in finance
- **arXiv (q-fin, cs.LG, stat.ML)** — preprints in quantitative finance and ML
- **Google Scholar** — supplementary search for grey literature and conference proceedings

Search queries combined financial terms with ML terms via boolean conjunction:

- **Financial terms:** “stock prediction”, “portfolio optimization”, “risk management”, “options pricing”, “market microstructure”, “quantitative finance”.
- **ML terms:** “machine learning”, “deep learning”, “neural network”, “random forest”, “gradient boosting”, “reinforcement learning”, “natural language processing”, “transformer”, “LSTM”.

The query was constructed as `(financial term) AND (ML term)` for every pair, with results aggregated and deduplicated across the five databases.

Initial searches were conducted in March 2026, with supplementary forward-citation searches completed in April 2026.

2.2 Inclusion and Exclusion Criteria

Table 1: Inclusion and exclusion criteria

Include	**Exclude**
Published 2015–2025	Published before 2015
Applies ML to a quantitative finance problem	Pure ML methodology with no financial application
Presents empirical results on financial data	Theoretical/conceptual only
Journal article, conference paper, or working paper	Theses, textbooks, blog posts
English language	Non-English
Uses real or realistic simulated financial data	Uses only toy/synthetic examples

2.3 Selection Process

1. **Identification:** Database searches returned 4,783 unique records after deduplication. The arXiv q-fin section alone contributed over 12,600 submissions during our review period (2015–2025), growing from 852 total submissions in 2015 to 2,435 in 2024—a nearly threefold increase reflecting the rapid growth of quantitative finance research (arXiv, 2025). Scopus, Web of Science, and SSRN contributed the remainder.

2. **Screening:** Title and abstract screening against inclusion criteria reduced the set to 847 papers.

3. **Eligibility:** Full-text review of screened papers identified 196 eligible studies.

4. **Inclusion:** Forward and backward citation searches added 31 additional papers, yielding a final sample of 227 studies.

2.4 Data Extraction Protocol

For each included study, we extracted the following fields into a structured database:

Table 2: Data extraction fields

Field	**Description**
Authors & Year	Standard citation information
Venue	Journal, conference, or preprint server
Application Domain	Return prediction, risk management, portfolio optimization, derivatives, microstructure, or alternative data
ML Method(s)	Primary and benchmark methods used
Asset Class	Equities, fixed income, FX, commodities, crypto, multi-asset
Data Source	Specific database or data provider
Sample Period	Start and end dates of empirical analysis
Geography	US, Europe, Asia, emerging markets, global
Primary Metric	Main performance measure reported
Key Finding	One-sentence summary of the main result
Code Available	Yes/No with URL if available
Data Available	Public / Proprietary / Mixed

2.5 Quality Assessment

We do not exclude studies based on quality scores, as methodological standards differ substantially between the ML and finance communities. Instead, we flag quality dimensions that readers should consider when interpreting results:

- **Out-of-sample testing:** Does the study use a proper temporal train/validation/test split, or does it employ cross-validation (which can introduce look-ahead bias in time series)?
- **Transaction costs:** Are trading strategies evaluated net of realistic transaction costs?
- **Benchmark comparison:** Is the ML model compared against appropriate traditional benchmarks (e.g., GARCH for volatility, Fama-French for returns)?
- **Statistical significance:** Are results tested for statistical significance, or are only point estimates reported?
- **Reproducibility:** Is code and/or data publicly available?

These dimensions are reported in our comparison tables (Appendix A) to help readers assess the strength of evidence for each finding.

3 A Taxonomy of Machine Learning in Finance

We organize the literature along two dimensions: *methodology* (the ML technique) and *application domain* (the financial problem). This section describes each dimension and presents the resulting taxonomy matrix.

3.1 Methodological Classification

We classify ML methods into five families, reflecting the major paradigms that have been applied in quantitative finance.

3.1.1 Tree-Based Ensemble Methods

Tree-based ensembles have emerged as the dominant methodology for structured (tabular) financial data. The family includes random forests (Breiman, 2001), gradient-boosted decision trees (Friedman, 2001), and their modern implementations: XGBoost (Chen and Guestrin, 2016), LightGBM (Ke et al., 2017), and CatBoost (Prokhorenkova et al., 2018). Their advantages in financial applications include:

- Natural handling of heterogeneous features (continuous, categorical, ordinal)
- Built-in feature importance measures that support interpretability
- Robustness to outliers and non-linear relationships
- Competitive performance without extensive hyperparameter tuning

- Ability to capture interaction effects without explicit feature engineering

The empirical dominance of GBDT in tabular data settings has been confirmed in large-scale benchmarks (Grinsztajn et al., 2022; Shwartz-Ziv and Armon, 2022), and financial applications consistently reflect this finding. In cross-sectional return prediction, GBDT methods match or exceed deep learning alternatives in the majority of studies reviewed (Section 4).

3.1.2 Deep Learning Architectures

Deep learning in finance has progressed through three distinct generations:

First generation (2015–2018): Recurrent networks. Long Short-Term Memory (LSTM) networks (Hochreiter and Schmidhuber, 1997) and Gated Recurrent Units (GRU) (Cho et al., 2014) were the first deep learning architectures widely applied to financial time series. Their ability to model sequential dependencies made them natural candidates for price prediction and volatility forecasting.

Second generation (2018–2022): Attention and transformers. The transformer architecture (Vaswani et al., 2017) and its variants have progressively displaced recurrent models. Financial applications include temporal fusion transformers (Lim et al., 2021), informer architectures for long-horizon forecasting (Zhou et al., 2021), and custom attention mechanisms for cross-sectional asset pricing (Chen et al., 2024).

Third generation (2022–2025): Foundation models. Pre-trained large language models (GPT, LLaMA, and domain-specific models such as FinBERT (Araci, 2019), BloombergGPT (Wu et al., 2023)) have opened new approaches to financial text analysis, event extraction, and even direct numerical forecasting through in-context learning.

Additional deep learning approaches applied in finance include convolutional neural networks (CNNs) for chart pattern recognition and volatility surface modeling, autoencoders for anomaly detection and factor extraction, and generative adversarial networks (GANs) for synthetic data generation and scenario simulation.

3.1.3 Reinforcement Learning

Reinforcement learning (RL) frames financial decision-making as a Markov Decision Process where an agent learns to take actions (trades) to maximize cumulative reward (risk-adjusted returns). Applications include:

- Portfolio allocation and rebalancing (Jiang et al., 2017; Ye et al., 2020)
- Optimal execution and market making (Ning et al., 2021)
- Options hedging (Buehler et al., 2019; Cao et al., 2021)

Key RL frameworks applied in finance include Deep Q-Networks (DQN), Proximal Policy Optimization (PPO), Soft Actor-Critic (SAC), and model-based approaches. A persistent challenge is the sim-to-real gap: policies learned in backtesting environments often fail to transfer to live markets due to non-stationarity and market impact.

3.1.4 Natural Language Processing and LLMs

NLP methods extract information from unstructured financial text: earnings calls, SEC filings, news articles, social media, and analyst reports. The field has evolved from bag-of-words and sentiment dictionaries (Loughran and McDonald, 2011) through word embeddings (Word2Vec, GloVe) to transformer-based models:

- **FinBERT** (Araci, 2019): BERT fine-tuned on financial text for sentiment classification
- **BloombergGPT** (Wu et al., 2023): 50B-parameter model trained on Bloomberg's financial data corpus
- **General-purpose LLMs**: GPT-4, Claude, and LLaMA applied zero-shot or few-shot to financial tasks

3.1.5 Hybrid and Ensemble Approaches

Many recent studies combine multiple methodologies:

- LSTM + attention mechanisms for enhanced sequence modeling
- NLP sentiment features fed into GBDT return prediction models
- RL agents using deep learning for state representation
- Stacking ensembles combining tree-based and neural network predictions

3.2 Application Domain Classification

We identify six application domains that collectively span the quantitative finance literature:

1. **Return prediction** (Section 4): Forecasting asset returns at various horizons (intraday to monthly), both cross-sectional (which stocks will outperform?) and time-series (will the market go up?).
2. **Risk management** (Section 5): Volatility forecasting, Value-at-Risk estimation, Expected Shortfall, credit risk modeling, and systemic risk measurement.

3. **Portfolio optimization** (Section 6): ML-enhanced portfolio construction, including direct weight learning, covariance estimation, and transaction cost optimization.

4. **Derivatives pricing and hedging** (Section 7): Option pricing beyond Black-Scholes, implied volatility surface modeling, and dynamic hedging strategies.

5. **Market microstructure** (Section 8): Order flow prediction, optimal execution, market making, and limit order book modeling.

6. **Alternative data and NLP** (Section 9): Extraction of tradeable signals from news, social media, satellite imagery, and other non-traditional data sources.

3.3 The Taxonomy Matrix

Table 3 presents the two-dimensional taxonomy, indicating the approximate number of studies in our sample at each intersection and our qualitative assessment of research maturity.

Table 3: Taxonomy matrix: Methods × Applications. Cell entries indicate approximate study count and maturity level (H = High, M = Medium, L = Low). Studies that apply more than one methodology (e.g., LSTM with attention, NLP features feeding a tree ensemble) appear in multiple rows; for this reason the cell totals (344) exceed the unique study count (227).

	Returns	**Risk**	**Portfolio**	**Derivatives**	**Microstr.**	**Alt. Data**
Tree-based	40+ (H)	15 (M)	10 (M)	5 (L)	5 (L)	10 (M)
Deep learning	35+ (H)	20 (M)	15 (M)	15 (M)	10 (M)	20 (M)
RL	5 (L)	3 (L)	20 (M)	10 (M)	8 (M)	2 (L)
NLP / LLM	15 (M)	5 (L)	3 (L)	2 (L)	2 (L)	30+ (H)
Hybrid	10 (M)	5 (L)	8 (M)	5 (L)	3 (L)	8 (M)

Several patterns emerge from this matrix:

- **Return prediction** is the most studied intersection, dominated by tree-based and deep learning methods.
- **RL for portfolio optimization** has attracted disproportionate attention relative to its empirical success, suggesting a gap between theoretical appeal and practical viability.
- **NLP/LLM methods beyond sentiment** (e.g., for risk management or derivatives) remain under-explored and represent opportunities for future work.

- **Market microstructure** is under-represented in the ML literature relative to its importance in practice, likely due to data access barriers.
- **Tree-based methods for derivatives** are surprisingly sparse, despite the tabular nature of options data.

4 Return Prediction

Return prediction is the oldest and most heavily studied application of ML in finance. We organize this section into cross-sectional prediction (which assets will outperform), time-series prediction (will the market go up), and the critical methodological considerations that determine whether reported results are reliable.

4.1 Cross-Sectional Return Prediction

The foundational work in ML-based cross-sectional prediction is Gu et al. (2020), who systematically compare linear models, tree-based methods, and neural networks for predicting the cross-section of monthly US stock returns. Their key findings have shaped the subsequent literature: neural networks and gradient-boosted trees significantly outperform linear models, and interaction effects between firm characteristics and macroeconomic variables matter.

4.1.1 The Tree-Based Dominance

Following Gu et al. (2020), numerous studies have confirmed that GBDT methods (XGBoost, LightGBM) deliver strong cross-sectional predictive performance:

- Leippold and Wang (2022) extend the analysis to international markets (30+ countries) and find that gradient-boosted trees generalize better across markets than deep learning methods, particularly in smaller markets with fewer training observations.
- Tobek and Hronec (2021) demonstrate that a substantial portion of the apparent predictability in the cross-section is absorbed by transaction costs, and that simpler models (including linear) close much of the gap with complex models after cost adjustment.
- Chen et al. (2024) introduce an attention-augmented GBDT that learns feature importance dynamically, improving on static tree-based models for return prediction across market regimes.

4.1.2 Deep Learning Approaches

Neural network approaches to cross-sectional prediction include:

- **Feed-forward networks:** Gu et al. (2020) establish baseline neural network performance. Subsequent work explores depth, regularization, and feature engineering choices.
- **Autoencoders for factor extraction:** Gu et al. (2021) propose conditional autoencoders that extract latent factors from firm characteristics, providing an ML-native alternative to Fama-French-style factor models. Kim et al. (2021) extend this to variational autoencoders.
- **Graph neural networks:** Chen et al. (2022) model the cross-section as a graph where stocks are nodes connected by industry, supply chain, or correlation edges, and use graph attention networks for prediction.
- **Transformers:** Duan and Zhang (2022) apply transformer architectures to the cross-section, treating each stock's characteristics as a token sequence. Cross-stock attention captures peer effects.

4.1.3 Comparison Table: Cross-Sectional Return Prediction

Table 4: Selected studies in cross-sectional return prediction

Study	Market	Period	Best Method	Metric	Code
Gu et al. (2020)	US	1957–2016	NN3	$R^2_{\mathrm{OOS}} = 0.40\%$	Yes
Leippold & Wang (2022)	Global	1990–2020	LightGBM	R^2_{OOS} varies	No
Tobek & Hronec (2021)	US	1963–2019	GBDT	Sharpe net of costs	Yes
Chen et al. (2024)	US	1970–2022	Attn-GBDT	$R^2_{\mathrm{OOS}} = 0.52\%$	No
Gu et al. (2021)	US	1967–2016	CA-AE	R^2_{OOS}	Yes
Duan et al. (2022)	US	1963–2021	Transformer	R^2_{OOS}	No

4.2 Time-Series Return Prediction

Time-series prediction (forecasting aggregate market or individual asset returns over time) faces the fundamental challenge that financial returns are close to unpredictable at short horizons (Welch and Goyal, 2008). ML methods have been applied to this problem with mixed results:

- **LSTM/GRU models:** Early deep learning studies (Fischer and Krauss, 2018; Bao et al., 2017) reported strong performance for individual stock prediction using LSTMs, but subsequent work has raised concerns about look-ahead bias and overfitting in these results.

- **Temporal Fusion Transformers:** Lim et al. (2021) propose the TFT architecture with variable selection, temporal attention, and quantile outputs, demonstrating state-of-the-art performance on financial forecasting benchmarks.
- **The efficient market baseline:** Avramov et al. (2023) argue that after proper regularization and economic constraints, the gains from ML over simple linear models for time-series prediction are modest, particularly at monthly and longer horizons. This finding is consistent with the adaptive markets hypothesis: predictability is time-varying and competed away.

4.3 Methodological Pitfalls in Return Prediction

A critical contribution of the recent literature is the identification of methodological pitfalls that inflate reported performance:

1. **Look-ahead bias:** Using future information in feature construction or normalization. Even subtle forms, such as using the full-sample mean for standardization, can materially affect results (de Prado, 2018).
2. **Survivorship bias:** Training only on stocks that survived to the end of the sample overstates predictability.
3. **Cross-validation in time series:** Standard k-fold cross-validation violates temporal ordering and leaks future information into training. Proper alternatives include expanding-window and rolling-window validation (Gu et al., 2020).
4. **Data snooping:** Testing many models and reporting only the best one inflates the probability of finding spurious predictability. Multiple testing corrections (e.g., Harvey et al. 2016) are essential but rarely applied.
5. **Ignoring transaction costs:** Strategies that appear profitable before costs may be worthless after realistic cost assumptions (Novy-Marx and Velikov, 2016; Tobek and Hronec, 2021).
6. **Short evaluation periods:** Strong performance during 2009–2020 (a prolonged bull market) does not generalize to different market regimes.

We recommend that future studies adopt the evaluation protocol of Gu et al. (2020) as a minimum standard: temporal splitting, economic metrics alongside statistical metrics, and comparison to appropriate benchmarks. Frankly, the fact that papers are still published in 2025 using k-fold cross-validation on time series data suggests that referee standards in some venues are too low.

5 Risk Management

ML applications in risk management span volatility forecasting, tail risk estimation, credit risk modeling, and systemic risk measurement. Unlike return prediction, where the signal-to-noise ratio is extremely low, risk management benefits from the persistence of volatility and the structure of tail dependencies, making it a domain where ML methods can deliver substantial improvements.

5.1 Volatility Forecasting

Volatility forecasting is the most mature ML application in risk management. The benchmark is the GARCH family (Bollerslev, 1986; Engle, 2002), and ML methods must demonstrate improvement over these well-established models.

5.1.1 Neural Network Approaches

- Bucci (2020) compare LSTM networks to GARCH, HAR-RV, and ARFIMA models for realized volatility forecasting on S&P 500 data. The LSTM improves upon GARCH models particularly during high-volatility regimes but shows marginal gains in calm markets.
- Zhang et al. (2019) propose a hybrid GARCH-LSTM model that uses GARCH residuals as LSTM inputs, combining the theoretical structure of GARCH with the flexibility of deep learning. This hybrid approach outperforms both standalone models.
- Christensen et al. (2023) apply temporal fusion transformers to multi-asset volatility forecasting and find that cross-asset attention (learning volatility spillovers) significantly improves forecasts for less liquid assets.

5.1.2 Tree-Based Approaches

- Luong and Dokuchaev (2021) show that random forests using a large set of lagged realized measures, option-implied features, and macroeconomic variables outperform HAR-RV models, particularly at weekly and monthly horizons.
- Mittnik et al. (2022) demonstrate that LightGBM achieves the best overall performance in a horse race of 15 volatility forecasting methods across 20 equity indices, with the advantage concentrated in forecasting regime transitions.

5.2 Value-at-Risk and Expected Shortfall

Traditional VaR models (historical simulation, parametric, Monte Carlo) often underestimate tail risk. ML approaches address this through:

- **Quantile regression forests** (Meinshausen, 2006): Non-parametric quantile estimation that naturally captures asymmetric tail behavior.
- **CAViaR extensions:** Taylor (2019) combine the Conditional Autoregressive VaR (CAViaR) framework with neural networks to model time-varying quantile dynamics.
- **GAN-based scenario generation:** Cont and Xu (2022) use conditional GANs to generate realistic tail scenarios for stress testing, outperforming historical simulation in terms of VaR backtest coverage.

5.3 Credit Risk

ML applications in credit risk include probability of default (PD) modeling, loss given default (LGD) estimation, and credit scoring. This sub-domain has the longest history of ML adoption in finance, predating the 2015 starting point of our review:

- Barboza et al. (2017) compare random forests, GBDT, and neural networks against logistic regression for bankruptcy prediction and find consistent improvement across all ML methods, with GBDT achieving the highest AUC.
- Bussmann et al. (2021) address the interpretability challenge by applying SHAP values to credit scoring models, enabling regulatory compliance while maintaining predictive power.
- **Alternative data:** Recent work incorporates satellite imagery, web traffic, and supply chain data into corporate credit models, bridging the credit risk and alternative data literatures (Suss and Trebbi, 2022).

5.4 Systemic Risk

ML approaches to systemic risk measurement include:

- **Network models:** Graph neural networks applied to interbank networks and correlation structures to identify systemically important institutions (Giudici and Polinesi, 2022).
- **Early warning systems:** Holopainen and Sarlin (2017) use random forests to predict banking crises, outperforming logistic regression early warning models used by central banks.
- **Contagion modeling:** Agent-based models enhanced with reinforcement learning to simulate crisis propagation (Delli Gatti and Grazzini, 2022).

5.5 Summary

Risk management is a domain where ML methods provide clear, economically significant improvements over traditional approaches. The persistence of volatility and the structured nature of risk factors create favorable conditions for ML. The key remaining challenge is regulatory acceptance: many risk management applications require model interpretability and stability that deep learning models do not naturally provide.

6 Portfolio Optimization

Traditional portfolio optimization, rooted in Markowitz mean-variance theory (Markowitz, 1952), suffers from well-documented estimation error problems: expected returns are notoriously difficult to estimate, and sample covariance matrices are unstable in high dimensions. ML offers two broad strategies: (i) improving the inputs to traditional optimization (better return and covariance estimates), and (ii) replacing optimization entirely with direct weight learning.

6.1 ML-Enhanced Inputs

6.1.1 Return Estimation

The return prediction methods reviewed in Section 4 directly improve portfolio inputs. Studies explicitly connecting prediction to portfolio performance include:

- Gu et al. (2020) show that portfolios sorted on neural network predicted returns earn significantly higher Sharpe ratios than traditional factor portfolios.
- DeMiguel et al. (2020) demonstrate that ML-predicted returns, combined with mean-variance optimization with short-sale constraints, outperform the equal-weight benchmark that has been notoriously difficult to beat.
- Cong et al. (2021) propose an asset pricing model where neural networks extract latent factors and estimate time-varying loadings simultaneously, yielding a factor model that is both economically interpretable and statistically superior.

6.1.2 Covariance Estimation

ML methods for covariance estimation address the curse of dimensionality:

- **Shrinkage via neural networks:** Ledoit and Wolf (2022) extend their influential shrinkage estimators with nonlinear shrinkage learned by neural networks, adapting to the eigenvalue distribution of the true covariance.

- **Factor models with ML:** Kelly et al. (2019) use instrumental PCA with neural network instruments to extract factors with time-varying loadings.

- **Graph-based structure:** Zhao et al. (2022) impose graph structure on the covariance matrix using sector and supply chain relationships, reducing estimation error in the off-diagonal elements.

6.2 Direct Weight Learning

An alternative paradigm bypasses explicit return and risk estimation, instead learning portfolio weights directly:

- Zhang et al. (2020a) train neural networks to output portfolio weights that maximize the Sharpe ratio end-to-end, avoiding the instabilities of two-step (predict-then-optimize) approaches.

- Butler and Kwon (2023) use differentiable convex optimization layers within neural networks, ensuring that output weights satisfy constraints (no short selling, maximum position sizes) by construction.

- Ban et al. (2018) propose a machine learning framework that simultaneously selects features and determines portfolio weights, demonstrating improved out-of-sample performance on US equity data.

6.3 Reinforcement Learning for Portfolio Management

RL is conceptually well-suited to portfolio management, as the sequential decision-making framework naturally captures multi-period rebalancing with transaction costs:

- Jiang et al. (2017) propose a CNN-based portfolio agent that learns from price images. While influential, the approach uses unrealistic assumptions (no transaction costs, unlimited liquidity).

- Ye et al. (2020) address this limitation with a cost-aware DRL agent that learns to trade-off rebalancing frequency against transaction costs.

- Wang et al. (2021) compare multiple RL algorithms (A2C, PPO, DDPG, TD3, SAC) in the FinRL framework, providing standardized baselines for RL portfolio management.

Our assessment is blunt: Despite the theoretical appeal, RL approaches to portfolio management have not convincingly demonstrated out-of-sample superiority over simpler alternatives. The number of papers proposing RL portfolio agents far exceeds the evidence that any of them work in practice. Key challenges include:

1. Non-stationarity of financial environments violates the stationary MDP assumption
2. Reward shaping (what risk-adjusted metric to maximize) significantly affects learned policies
3. Sample inefficiency: RL requires many episodes, but financial history provides limited independent samples
4. The sim-to-real gap: policies learned in backtests often degrade in live trading

6.4 Transaction Cost Integration

A distinguishing feature of modern ML portfolio methods is explicit transaction cost integration:

- DeMiguel et al. (2020) impose ℓ_1 turnover constraints that simultaneously regularize the portfolio and control costs.
- Moallemi and Saglam (2023) learn a turnover-optimal trading policy that balances the tracking error of a target portfolio against trading costs.
- Multi-period approaches using dynamic programming or RL naturally incorporate path-dependent costs, including market impact models (Almgren and Chriss, 2001).

7 Derivatives Pricing and Hedging

Derivatives pricing presents a unique setting for ML: unlike return prediction, where the signal-to-noise ratio is extremely low, options pricing involves learning a structured function (the pricing map from inputs to fair value) that is theoretically well-understood but difficult to compute for complex products. This makes it a natural domain for function approximation via neural networks.

7.1 Option Pricing

7.1.1 Neural Network Pricing Models

The idea of using neural networks as function approximators for option pricing dates to Hutchinson et al. (1994), but modern deep learning has substantially expanded the scope:

- Bayer and Stemper (2019) train deep neural networks to approximate the pricing function for rough volatility models, achieving computation times orders of magnitude faster than Monte Carlo while maintaining accuracy.

- Horvath et al. (2021) extend this to general stochastic volatility models, showing that neural network approximation of the pricing map is a viable alternative to finite-difference PDE solvers for complex payoffs.
- Ruf and Wang (2020) compare neural network approaches to option pricing across data-driven models (trained on market data), model-driven approaches (trained on model outputs), and hybrid methods. They find that data-driven approaches outperform parametric models for plain vanilla options but struggle with illiquid exotics.

7.1.2 Implied Volatility Surface Modeling

The implied volatility surface, the function mapping strike and maturity to Black-Scholes implied volatility, is a central object in options markets:

- Ackerer and Filipović (2020) propose a neural network parameterization of the implied volatility surface that guarantees absence of calendar and butterfly arbitrage by construction.
- Cohen et al. (2023) use variational autoencoders to learn a low-dimensional representation of the vol surface, enabling surface interpolation, extrapolation, and scenario generation.
- **SVI and beyond:** The Stochastic Volatility Inspired (SVI) parameterization (Gatheral and Jacquier, 2014) remains the industry benchmark. ML approaches must be compared against this well-calibrated parametric model, and gains are typically in the tails and short maturities.

7.2 Dynamic Hedging

ML-based hedging strategies learn to minimize hedging error directly, without assuming a specific pricing model:

- Buehler et al. (2019) introduce the "deep hedging" framework: a neural network learns the optimal hedging strategy by minimizing a convex risk measure of the hedging error. This approach naturally incorporates transaction costs, market frictions, and model uncertainty.
- Cao et al. (2021) apply reinforcement learning to delta hedging, with the agent learning to adjust hedge ratios in response to market conditions. The RL approach outperforms delta-gamma hedging in the presence of jumps and stochastic volatility.
- Murray et al. (2022) extend deep hedging to incorporate multiple hedging instruments (options at various strikes) and show that the learned strategy effectively rediscovers vega and gamma hedging without being told about the Greeks.

7.3 Exotic Derivatives and Structured Products

For complex derivatives where analytical pricing is unavailable:

- **American options:** Becker et al. (2019) propose a deep learning approach to American option pricing that scales to high dimensions (baskets of 100+ underlyings), overcoming the curse of dimensionality that limits traditional Longstaff-Schwartz regression.
- **Callable structures:** Neural networks approximate the continuation value function in backward induction, enabling real-time pricing of callable bonds and structured notes.
- **XVA computation:** Gnoatto et al. (2023) apply neural network regression to credit valuation adjustments (CVA, DVA, FVA), achieving massive speedups over nested Monte Carlo.

7.4 Summary

In our view, derivatives pricing is the most genuinely successful application domain for deep learning in finance, and it is telling that it gets less attention than return prediction. The reasons for its success are structural: (i) the target function is smooth and structured, (ii) training data can be generated synthetically from pricing models, (iii) the accuracy requirements are well-defined, and (iv) the computational savings are economically significant. Deep hedging marks a clean break from model-based hedging and is increasingly adopted in practice.

8 Market Microstructure

Market microstructure, the study of trading mechanisms, price formation, and liquidity, generates some of the most data-rich problems in finance. Limit order books (LOBs) produce millions of events per day per instrument, creating both opportunities and challenges for ML methods.

8.1 Limit Order Book Modeling

8.1.1 Mid-Price Prediction

Short-horizon mid-price prediction from LOB data is a canonical microstructure ML task:

- Sirignano and Cont (2019) train deep learning models on LOB data from hundreds of stocks simultaneously, finding that a universal model (trained across stocks) outperforms stock-specific models, suggesting common microstructure patterns across assets.

- Zhang et al. (2020b) apply LSTMs to LOB snapshots and demonstrate that temporal patterns in the order flow carry predictive information beyond the current book state.
- Kolm et al. (2023) benchmark CNNs, LSTMs, transformers, and tree-based methods on a standardized LOB dataset (FI-2010, Ntakaris et al. 2018), finding that transformers with positional encoding achieve the best classification accuracy for directional prediction at horizons of 10–100 events.

8.1.2 Full Distribution Modeling

Beyond point prediction, some studies model the full distribution of LOB evolution:

- Cont and Kalinin (2021) use neural point processes to model the arrival of limit orders, market orders, and cancellations as a marked point process, capturing the clustering and self-exciting nature of order flow.
- Shi et al. (2022) apply normalizing flows to model the joint distribution of multi-level LOB states, enabling realistic scenario generation for execution simulation.

8.2 Optimal Execution

Optimal execution, minimizing the market impact of large orders, is a natural RL application:

- Ning et al. (2021) compare RL-based execution strategies (DQN, PPO) against the Almgren-Chriss analytical benchmark (Almgren and Chriss, 2001) and find that RL agents learn to exploit intraday liquidity patterns, achieving lower implementation shortfall.
- Fang et al. (2021) use multi-agent RL to model the interaction between a large executor and adaptive market makers, capturing the game-theoretic aspects of execution that single-agent models miss.
- Schnaubelt (2022) train a meta-learner that adapts execution strategies to changing market conditions (volatility regimes, liquidity shocks) without retraining.

8.3 Market Making

Automated market making, providing liquidity by continuously quoting bid and ask prices, has been extensively studied with RL:

- Guéant (2017) provide the theoretical foundation: Avellaneda-Stoikov style models with analytical solutions that serve as benchmarks for ML approaches.

- Spooner et al. (2018) apply DRL (DQN and variants) to market making in a realistic LOB simulator, showing that the learned policy adapts spread and inventory management to market conditions.

- Gašperov and Kostanjcar (2021) propose a multi-agent market making framework where multiple RL agents interact in a simulated market, producing emergent dynamics (bid-ask bounce, adverse selection) consistent with empirical microstructure.

8.4 Data and Access Challenges

A distinctive feature of the microstructure literature is the data access barrier. Tick-level LOB data is expensive and often proprietary:

- The FI-2010 dataset (Ntakaris et al., 2018) (Helsinki Stock Exchange, 10 stocks, 10 business days) has become a de facto standard, but its small size and non-US market limit generalizability.

- LOBSTER (NASDAQ data) is widely used in academic studies but requires institutional subscription.

- Crypto exchanges (Binance, Coinbase) provide free LOB data, making cryptocurrency microstructure an increasingly popular testbed.

This data bottleneck explains the odd situation where microstructure, a domain that generates more data per day than any other in finance, is under-represented in the ML literature. Researchers who want to work on LOB modeling face a choice between the small, dated FI-2010 dataset and expensive commercial data. The growing availability of free crypto exchange data may eventually break this logjam, though cryptocurrency market microstructure has its own peculiarities.

9 Alternative Data and Natural Language Processing

Alternative data (non-traditional sources including news, social media, satellite imagery, web traffic, and corporate filings) represents the fastest-growing application area for ML in finance. NLP methods are the primary tool for extracting signals from textual alternative data.

9.1 Financial Sentiment Analysis

9.1.1 Dictionary-Based Methods

The foundation of financial text analysis is the Loughran and McDonald (2011) dictionary, which demonstrated that general-purpose sentiment dictionaries (e.g., Harvard General Inquirer) perform poorly on financial text because words like "liability," "tax," and "cost" have neutral meanings in financial context. The Loughran-McDonald dictionary remains a benchmark against which ML approaches are evaluated.

9.1.2 Pre-trained Language Models

The transformer revolution has reached financial NLP:

- **FinBERT** (Araci, 2019): BERT fine-tuned on financial news. Achieves substantially higher accuracy than dictionary methods and general-purpose BERT on financial sentiment classification.
- **BloombergGPT** (Wu et al., 2023): A 50-billion parameter LLM trained on Bloomberg's proprietary corpus of financial documents. Outperforms general-purpose LLMs on financial NLP benchmarks while maintaining competitive general capabilities.
- **General LLMs (GPT-4, Claude):** Lopez-Lira and Tang (2023) show that GPT-3.5/4 can predict stock returns from news headlines, with performance comparable to purpose-trained models. The zero-shot capability eliminates the need for labeled training data.
- **Open-source finance LLMs:** FinGPT (Yang et al., 2023) and InvestLM provide open-source alternatives for financial NLP, enabling reproducible research.

9.2 News and Event Analysis

Beyond sentiment, ML methods extract structured information from news:

- **Event detection:** Identifying market-moving events (M&A announcements, earnings surprises, regulatory changes) from news feeds in real-time (Ein-Dor et al., 2020).
- **Causal extraction:** Extracting causal relationships ("tariffs caused a decline in imports") from financial news using transformer-based information extraction (Mariko et al., 2022).
- **Earnings call analysis:** ML models analyze the text and audio of earnings calls, extracting CEO tone, uncertainty language, and deceptive patterns that predict post-call stock movements (Li et al., 2021; Qin and Yang, 2019).

9.3 Social Media and Crowd Signals

- **Twitter/X:** Bollen et al. (2011) pioneered Twitter mood prediction of DJIA movements. More recent work uses transformer models on high-frequency tweet streams, with mixed evidence of genuine predictive content versus noise (Renault, 2020).
- **Reddit (WallStreetBets):** The GameStop episode (January 2021) catalyzed academic interest in retail investor coordination on social media. Bradley et al. (2021) analyze the information content of Reddit posts for stock prediction.
- **StockTwits:** A dedicated financial social platform that provides labeled bullish/bearish sentiment, making it a convenient training dataset for sentiment models (Oliveira et al., 2017).

9.4 Non-Textual Alternative Data

- **Satellite imagery:** CNNs applied to satellite images of parking lots (retail foot traffic), oil storage tanks (inventory levels), and agricultural fields (crop yields). Katona et al. (2021) demonstrate that satellite-derived parking lot occupancy predicts retailer revenue ahead of official reports.
- **Web data:** Job postings, product reviews, app download statistics, and web traffic as leading indicators of corporate performance (Green et al., 2019).
- **ESG and climate:** NLP extraction of ESG metrics from corporate reports, news, and regulatory filings. This subfield is growing rapidly due to regulatory pressure for ESG disclosure (Kalesnik and Luo, 2022).

9.5 Challenges and Limitations

1. **Signal decay:** Alternative data signals are rapidly incorporated into prices once widely adopted, creating an arms race for novel data sources.
2. **Backtest inflation:** Alternative data is often available only for recent periods, limiting out-of-sample evaluation.
3. **Labeling ambiguity:** Financial sentiment is context-dependent (“interest rates rose” is positive for banks, negative for borrowers), making supervised training challenging.
4. **Cost:** Commercial alternative data is expensive, creating barriers to academic research and replication.

10 Cross-Cutting Themes

The domain-specific sections above tell individual stories, but several themes recur often enough to warrant separate treatment. We discuss five: the interpretability problem, the non-stationarity problem, the compute landscape, the field's most serious issue (the reproducibility crisis), and a measurement instrument we propose to track it (the Reproducibility Disclosure Score).

10.1 The Interpretability Problem

Finance has a regulatory environment that other ML application domains do not. A bank deploying a credit scoring model in the EU must explain, under GDPR, why an applicant was denied. A portfolio manager who tells clients "the neural network said so" will not retain those clients for long. This creates genuine tension: the methods that predict best are often the hardest to explain.

The standard response has been post-hoc explanation tools. SHAP (Lundberg and Lee, 2017) has become nearly ubiquitous in financial ML papers; our sample includes over 30 studies that use it. The appeal is obvious: Shapley values provide a theoretically grounded decomposition of any model's prediction into feature contributions. Bussmann et al. (2021) demonstrate this for credit scoring; Leippold and Wang (2022) apply it to return prediction models across international markets.

But there is a problem that most papers gloss over. SHAP explanations for tree ensembles are exact, but SHAP explanations for deep networks rely on approximations (KernelSHAP, DeepSHAP) that can be unstable: two runs on the same model can produce different feature rankings. Ribeiro et al. (2016)'s LIME has similar instability issues. In practice, the "explanations" these tools generate are often post-hoc rationalizations rather than genuine insights into model behavior, and the finance literature has been slow to reckon with this.

A more promising direction, in our assessment, is building interpretability into the model from the start. Chang et al. (2021) apply neural additive models that are inherently additive, with each feature's contribution plotted independently. The accuracy tradeoff is smaller than one might expect, particularly for the cross-sectional prediction tasks where tabular data dominates.

10.2 Non-Stationarity: The Fundamental Challenge

If there is one thing that makes financial ML genuinely harder than, say, image classification, it is non-stationarity. The data-generating process shifts constantly, driven by regime changes, central bank interventions, market structure evolution, and (critically) the fact that other participants adapt. A signal that predicts returns today gets traded on, moves prices, and weakens the very pattern it exploited. McLean and Pontiff (2016) documented this phenomenon empirically: anomalies published in academic papers lose roughly half their predictive power in the three years following publication.

This creates a paradox for ML models. More complex models can fit more patterns, but in a non-stationary environment, many of those patterns are ephemeral. A model that achieves excellent in-sample R^2 by memorizing regime-specific relationships will fail catastrophically when the regime changes. This likely explains a finding that runs through the literature like a thread: simpler models (especially linear models with strong regularization) often match or beat deep learning out of sample, particularly at longer horizons (Avramov et al., 2023).

The field has proposed several responses. Huck (2019) advocate online learning frameworks that update model parameters with each new observation. Nystrup et al. (2020) train separate models for each market regime and use a detection layer to switch between them. Transfer learning (pre-training on long historical samples and fine-tuning on recent data) shows promise in some settings (Li et al., 2022). But none of these approaches solves the problem fully, and we are skeptical that any purely statistical method can, because the non-stationarity is not statistical noise; it is generated by adaptive, strategic agents who are themselves trying to predict the market.

10.3 Compute: Not the Bottleneck You Think

It is tempting to assume that the main computational challenge in financial ML is training large models. It is not. The binding constraint for most researchers is backtesting. A responsible backtest requires rolling-window retraining: fit a model on data up to time t, predict time $t + 1$, advance the window, and repeat. For a cross-sectional model over 60 years of monthly data with 3,000 features and 5,000 stocks, this means fitting hundreds of models, each on a large dataset. de Prado (2018)'s combinatorial purged cross-validation, while statistically superior to simple time-series splits, multiplies this cost further.

On the hardware side, the picture has changed dramatically. Consumer-grade Apple Silicon (M-series) now provides 128 GB of unified memory and 40 GPU/CPU cores in a laptop. Cloud compute costs have fallen to the point where a large-scale backtest on AWS runs in hours for under $100. These developments have meaningfully democratized financial ML research: computations that required institutional infrastructure five years ago can now be done by a solo researcher with a good laptop.

That said, inference latency still matters for deployment. Models destined for intraday trading face hard constraints: a gradient-boosted tree can score a universe of 3,000 stocks in milliseconds, while a transformer with cross-stock attention may take seconds. There is an underappreciated design tradeoff here: the marginal accuracy gain from a complex model must be weighed against the latency cost, and in most practical settings, the simpler model wins.

10.4 The Reproducibility Crisis

We saved this for last because we consider it the most important issue in the field. Financial ML has a reproducibility problem that is, if anything, worse than the broader ML reproducibility crisis, and that crisis is already severe.

The numbers from our survey are sobering. Only approximately 21% of the 227 studies in our sample provide source code, consistent with findings across top-tier ML venues (Pineau et al., 2021). Papers that share code on GitHub receive roughly 20% more citations per month (Hasselbring et al., 2024), so there is a clear individual incentive to share, yet most authors do not.

Code availability is only part of the problem. Consider what it takes to replicate a financial ML study:

1. **Data.** Many studies use CRSP, Compustat, or OptionMetrics: databases that cost thousands of dollars per year. A researcher at a university without these subscriptions simply cannot attempt replication. Others use proprietary Bloomberg data that is contractually impossible to share.

2. **Preprocessing.** Financial data requires extensive cleaning: handling delisted stocks, adjusting for splits and dividends, merging across databases with different identifier systems (CUSIP, PERMNO, GVKEY). These choices materially affect results, and papers rarely document them in sufficient detail.

3. **Hyperparameters.** Two implementations of "XGBoost with 500 trees" can produce different results depending on learning rate, regularization, subsampling, early stopping criteria, and random seed. Most papers report only a subset of these choices.

4. **Selective reporting.** Nobody publishes a paper titled "We tried deep learning on stock returns and it didn't work." The publication filter means the literature overstates the average efficacy of ML methods. We have no way to estimate the magnitude of this bias, but the fact that every paper in our sample reports positive results for at least one ML method, despite the efficient market hypothesis suggesting that persistent predictability should be hard to find, is suggestive.

The field needs a cultural shift toward mandatory code sharing and standardized benchmarks. The computer vision community solved an analogous problem with ImageNet; quantitative finance needs its equivalent. Recent efforts like FinRL (Wang et al., 2021) and the Tidy Finance replication project are steps in the right direction, but they remain grassroots efforts rather than community standards.

This survey's reproducibility commitment. A literature review that critiques the field's reproducibility while sharing none of its own data would be self-defeating. Accordingly, the full annotated 227-study database underlying this survey is released as a CSV at `https://github.com/ayk5511/ml-finance-survey`, together with the search query log, the screening decisions, and the LaTeX source of this paper. For each study the database records authors, year, venue, application domain, primary ML method, benchmark, asset class, sample period, geography, primary metric, key finding, code-availability flag, and data-availability flag. We invite readers to flag missing studies, miscategorizations, or incorrect code/data flags via GitHub Issues; corrections will be incorporated in subsequent revisions.

10.5 The Reproducibility Disclosure Score

The qualitative critique above is hard to act on. To turn it into something measurable, we propose a simple Reproducibility Disclosure Score (RDS) on the integer scale $\{0, 1, 2\}$:

- +1 point if replication code is publicly available (typically at GitHub, Zenodo, or a journal's supplementary site).
- +1 point if the data underlying the headline result is openly accessible (no paywall, no institutional credential required) or fully simulated from a generative model documented in the paper.

The RDS is deliberately conservative: it captures *open materials*, not whether those materials are sufficient for end-to-end replication. A study with public code and public data still scores 2 even if its hyperparameters or preprocessing pipeline are under-documented. We see this as a strength rather than a weakness for a first-generation rubric. Scoring open materials is objective and verifiable from the paper alone, whereas scoring true replicability would require attempting to replicate each study, which is infeasible at survey scale. Future versions of the rubric might add subscores for hyperparameter completeness, preprocessing transparency, and verified replication.

We applied the RDS to the 99 empirical studies in our database for which the rubric is meaningful, excluding 20 foundational, baseline, or methodology papers where code and data availability are not the right questions (e.g., Markowitz 1952, Bollerslev 1986, Vaswani et al. 2017). Table 5 reports the distribution overall and across three cuts.

Table 5: Reproducibility Disclosure Score across 99 empirical studies in the survey database. Mean RDS is 0.78 on a 0–2 scale; only 16% of studies disclose both code and public-or-simulated data. Per-study scores are released in `data/survey_database.csv`.

Cut	**N**	**Mean RDS**	**% scoring 2**
Overall	99	0.78	16%
By application domain			
Microstructure	11	0.91	27%
Risk management	14	0.86	0%
Derivatives	12	0.75	17%
Alternative data & NLP	14	0.71	14%
Return prediction	26	0.69	19%
Portfolio optimization	10	0.50	20%
By venue type			
Preprints (arXiv, SSRN, working)	11	0.91	36%
Journals and conferences	88	0.76	14%
By era			
2008–2019	36	0.89	19%
2020–2024	63	0.71	14%

Three findings stand out. First, only 16% of empirical studies score the maximum 2. The remaining 84% require either a paid data subscription or unreleased code to reproduce, before we even consider finer-grained issues like hyperparameter documentation. Second, the rate of full disclosure has *fallen* between the early and recent periods (19% in 2008–2019 versus 14% in 2020–2024), driven by an influx of recent studies using proprietary alternative data and proprietary LLM APIs. Third, preprints score higher than peer-reviewed venues (mean 0.91 versus 0.76), which is consistent with preprint authors having stronger incentives to demonstrate replicability in lieu of formal peer review. Risk management is a notable outlier: a relatively high mean RDS (0.86) but zero studies scoring the maximum 2, because credit and banking studies use real institutional data that is never public even when the code is.

The per-study RDS is recorded in the supplementary database, which lets readers filter the literature by reproducibility threshold or replicate the cuts above. We encourage other survey authors to apply the same rubric to their own samples; a comparable score across surveys would let the community track whether financial ML reproducibility is improving or deteriorating over time.

11 Open Problems and Research Agenda

Every survey ends with a section on "future directions," and most such sections are forgettable wishlists. We will try to be more specific. What follows are ten problems that are both tractable and important, ranked roughly by how much impact a solution would have on the field.

11.1 Problem 1: Build the Financial ImageNet

Computer vision took off when ImageNet gave everyone a common benchmark to compete on. Financial ML has no equivalent. Each paper defines its own universe, its own features, its own evaluation window. The result is that we cannot meaningfully compare across studies: did paper A beat paper B's method, or did it just test on an easier dataset?

What the field needs is one or more open benchmark datasets with standardized features, train/test splits, and evaluation metrics. The FinRL library (Wang et al., 2021) and Tidy Finance replication project are early attempts, but they are grassroots efforts, not community standards. This is the single highest-leverage intervention anyone could make for the field. It is also not glamorous, which is why nobody has done it yet.

11.2 Problem 2: The Backtest-to-Live Gap

Every practitioner knows that backtests lie. Models that earn beautiful Sharpe ratios on historical data reliably disappoint when deployed. The causes are well-known (market impact, latency, slippage, data snooping, non-stationarity), but we lack good frameworks for *quantifying* the expected degradation before going live.

Bailey and de Prado (2014) propose minimum backtest length criteria and de Prado (2018) develop combinatorial purged cross-validation, but these are patches, not solutions. A serious research program here would study the gap empirically, using live trading records from firms willing to share them (admittedly a hard ask), and develop prediction intervals around backtest performance that account for the known sources of inflation.

11.3 Problem 3: Models That Know When the World Changed

Non-stationarity is the fundamental difficulty of financial ML (see Section 10), and the field's response so far has been improvised rather than principled. Rolling windows, regime-switching models, and transfer learning each address one facet of the problem. What is missing is a framework that treats non-stationarity not as a nuisance parameter but as the core modeling challenge.

Meta-learning, learning to learn from few samples in new environments, is a promising direction that has barely been explored in finance. So is the idea that model *simplicity* might be a feature rather than a bug in non-stationary environments: a linear model with five features may degrade more gracefully than a deep network with five hundred.

11.4 Problem 4: What Exactly Should a Financial Foundation Model Look Like?

BloombergGPT (Wu et al., 2023) showed that a large language model trained on financial text can outperform general-purpose LLMs on financial NLP tasks. But the deeper question is whether this approach makes sense at all. Financial data is fundamentally multimodal. It includes time series, cross-sectional features, tabular data, text, and (increasingly) images and audio. An LLM trained only on text is leaving most of the information on the table.

The interesting research question is whether someone can build a foundation model that ingests all these modalities jointly. The closest analogy outside finance might be the time-series foundation models (TimesFM, Chronos) emerging from Google and Amazon, but those handle only univariate series. A genuinely useful financial foundation model would need to handle panel data, text, and time series simultaneously, and nobody has demonstrated that this is feasible.

11.5 Problem 5: Causation, Not Just Correlation

Almost every paper in our survey trains a model to predict Y from X. Almost none ask whether X *causes* Y. This matters for two reasons. First, causal features should be more stable across regimes than merely correlated features. Second, policy questions (what happens if the Fed raises rates? what if a new regulation is imposed?) are inherently causal and cannot be answered by predictive models.

The econometrics community has developed powerful causal inference tools (double ML (Chernozhukov et al., 2018), causal forests, synthetic controls), but these have barely penetrated the financial ML literature. Bridging these communities is an obvious opportunity.

11.6 Problem 6: Markets Are Games, Not Datasets

Standard ML treats financial data as fixed: here is a dataset, learn a function, evaluate on a test set. But financial markets are multi-agent games where your model's predictions, once acted upon, change the very data your model was trained on. This is not an abstract concern; it is the mechanism by which alpha decays.

Multi-agent reinforcement learning offers a framework for thinking about this, and a few papers have explored it for market making and execution. But the harder problem (modeling how the aggregate behavior of many ML-equipped agents reshapes return distributions) remains wide open. The flash crash literature suggests this is practically important, not just theoretically interesting.

11.7 Problem 7: Algorithmic Fairness in Lending and Insurance

Credit scoring and insurance pricing are the two financial applications where ML bias has regulatory teeth. The EU's AI Act and the US CFPB's enforcement actions have made this an urgent practical problem, not just an academic one. The core difficulty is that proxy discrimination, where ML models learn to use zip codes or shopping patterns as proxies for race, is hard to detect and harder to prevent without sacrificing predictive accuracy.

11.8 Problem 8: Learning Without Sharing Data

Banks have data that, pooled together, would dramatically improve credit risk models, fraud detection, and systemic risk monitoring. They will never pool it directly. Federated learning, where each institution trains locally and shares only model updates, is the obvious technical solution, but financial applications add complications: heterogeneous data distributions across banks, regulatory constraints on even model parameters, and the incentive problem that the best-data bank gains the least from federation.

11.9 Problem 9: Climate Risk Quantification

Climate risk is the newest frontier for financial ML, driven by regulatory mandates (TCFD, EU taxonomy) rather than alpha-seeking. The key ML applications are NLP extraction of climate risk disclosures from corporate filings, integration of physical risk models with asset valuations, and carbon-intensity factor construction. The data infrastructure is immature, which means there is a first-mover advantage for researchers who build good datasets now.

11.10 Problem 10: Audit Trails for Regulator-Defensible AI

The Federal Reserve's SR 11-7 guidance, the OCC's 2011-12 model risk management bulletin, and the EU AI Act each impose what amounts to the same requirement on financial institutions deploying ML: the entire model-development lifecycle must be auditable, including data provenance, hyperparameter search, model selection, and post-deployment drift monitoring. Standard ML practice, including the practice in most papers we surveyed, leaves few audit-grade artifacts. A typical paper reports the final model and its hyperparameters; it does not record the dozen architectures that were tried and discarded, the random seeds that were used to seed each search, or the backtest variants considered before settling on the published one.

This gap is not closed by current MLOps tooling. Frameworks such as MLflow and Weights & Biases track operational metrics like loss curves and validation accuracy, but were not designed to produce tamper-evident records suitable for a regulator's third-line-of-defense audit. The open research problem is to build training and deployment pipelines that emit cryptographically chained, append-only logs of every model-relevant decision, and to define a minimum schema such logs must satisfy to count as audit-grade. The closest analog outside finance is supply-chain attestation work like SLSA and in-toto; bringing that rigor to ML model development would address a real, regulator-driven need that the survey literature has barely engaged with.

12 Conclusion

Having worked through 227 studies across six domains, we step back and offer the assessment we wish someone had given us before we started this review.

Trees still win on tabular data. This is the single most consistently replicated finding in the literature, and it surprises people who equate "machine learning" with deep learning. For cross-sectional return prediction, the most studied problem in financial ML, gradient-boosted trees (XGBoost, LightGBM) match or beat neural networks in the clear majority of papers. This is consistent with evidence outside finance (Grinsztajn et al., 2022), and it has practical implications: tree models are faster to train, easier to interpret, and less sensitive to hyperparameters than deep networks. If you are a practitioner deploying your first ML model for stock selection, start with LightGBM, not a transformer.

Transformers have displaced LSTMs. Within the subset of problems where deep learning does add value (long-horizon time series, order flow modeling, NLP), the attention mechanism has won. We found no study published after 2022 where an LSTM outperformed a transformer variant in a head-to-head comparison. The transition happened faster in finance than in most other applied domains, probably because the finance community was already primed by the NLP community's results.

The best results are in risk and derivatives, not return prediction. This may be the most underappreciated finding. Return prediction attracts the most papers, but the highest ratio of genuine improvement over traditional methods is in volatility forecasting (ML vs. GARCH), derivatives pricing (neural networks vs. Monte Carlo), and hedging (deep hedging vs. delta-gamma). The reason is structural: returns have a terrible signal-to-noise ratio, while volatility is persistent and pricing maps are smooth functions. If the goal is to find problems where ML offers clear, economically meaningful improvements, risk management and derivatives are where to look.

LLMs are moving fast, but evidence is thin. The progression from sentiment dictionaries to FinBERT to GPT-4 is real, and Lopez-Lira and Tang (2023) show that zero-shot GPT predictions have genuine information content for stock returns. But most LLM-in-finance papers have appeared as preprints, use short sample periods (often starting in 2022), and have not been independently replicated. We are cautiously optimistic about this direction but wary of the hype cycle. The claim that a general-purpose LLM can replace domain-specific models is, at best, premature.

Reproducibility is the field's biggest problem. We keep coming back to this. Only 21% code sharing. Proprietary data in the majority of studies. Selective reporting of positive results. These are not minor methodological quibbles; they mean that we cannot be confident in the literature's overall conclusions. A field that cannot reproduce its own results cannot credibly claim to have established much of anything. Until this changes, every claim about ML "beating" traditional methods in finance should be read with a healthy dose of skepticism.

Looking ahead, we see the highest-impact opportunities in three areas: (1) foundation models specifically designed for financial data, combining numerical and textual inputs in ways that current architectures do not; (2) causal ML methods that go beyond prediction to answer counterfactual questions about markets and policy; and (3) the basic infrastructure work of building standardized benchmarks and reproducibility standards that the field badly needs. The first two are intellectually exciting. The third is less glamorous but, in our view, more important.

Replication materials. The full annotated database of all 227 studies surveyed here, the LaTeX source for this paper, and supplementary scripts are available at https://github.com/ayk5511/ml-finance-survey under permissive licenses (CC-BY 4.0 for text, MIT for code). Readers are encouraged to use the database as a starting point for their own systematic reviews, to file corrections via GitHub Issues, and to fork the repository to extend coverage into new subfields.

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A Selected Representative Studies, by Domain

These tables present a curated sample of representative studies in each of the six application domains, chosen to span the methodological families discussed in Section 3. They are not the full 227-study sample. The complete annotated database (covering every included study with its venue, asset class, sample period, ML method, benchmark, primary metric, key finding, and code/data availability) is released as a CSV in the supplementary repository at `https://github.com/ayk5511/ml-finance-survey` (`data/survey_database.csv`).

For each tabulated study we report: authors and year, asset class, sample period, primary ML method, primary benchmark, headline performance metric, code availability (Y/N), and data source. The Code column reflects whether replication code is publicly available at the time of writing; "–" indicates the field is not applicable (e.g., theoretical or simulation-only studies). Asset and period entries shown as "–" indicate either non-empirical studies or study designs where the field does not apply.

Table 6: Representative studies: Return Prediction.

Study	Year	Asset	Period	ML Method	Bench.	Metric	Code	Data
Welch and Goyal (2008)	2008	US Eq.	1927–2005	Linear baseline	Hist. mean	R^2_{OOS}	Y	Public
Bao et al. (2017)	2017	US Eq.	2007–2016	SAE+LSTM	ARIMA	MAPE	N	Yahoo
Fischer and Krauss (2018)	2018	US Eq.	1992–2015	LSTM	RF, Logit	Accuracy	Y	Public

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Table 6 continued

Study	Year	Asset	Period	ML Method	Bench.	Metric	Code	Data
Kelly et al. (2019)	2019	US Eq.	1962–2014	IPCA	PCA, FF5	R^2_{OOS}	N	CRSP
Gu et al. (2020)	2020	US Eq.	1957–2016	NN, RF, GBDT	OLS, LASSO	R^2_{OOS}	Y	CRSP
DeMiguel et al. (2020)	2020	US Eq.	1980–2014	LASSO + GBDT	FF5, EW	Sharpe	N	CRSP
Feng et al. (2020)	2020	US Eq.	1980–2016	LASSO (t-test)	150 factors	SDF R^2	N	CRSP
Freyberger et al. (2020)	2020	US Eq.	1965–2014	Adaptive LASSO	Linear	R^2_{OOS}	Y	CRSP
Kozak et al. (2020)	2020	US Eq.	1973–2017	Elastic Net	FF, PCA	SDF R^2	N	CRSP
Cong et al. (2021)	2021	US Eq.	1963–2019	Panel Tree	FF5	R^2_{OOS}	N	CRSP
Gu et al. (2021)	2021	US Eq.	1967–2016	Cond. Autoenc.	PCA	R^2_{OOS}	Y	CRSP
Kim et al. (2021)	2021	US Eq.	1965–2017	VAE factors	PCA, IPCA	R^2_{OOS}	N	CRSP
Tobek and Hronec (2021)	2021	US Eq.	1963–2019	GBDT (cost-aware)	Fama-French	Sharpe (net)	Y	CRSP
Chen et al. (2022)	2022	US Eq.	–	Graph NN	Linear, NN	IC, accuracy	N	Public
Duan and Zhang (2022)	2022	US Eq.	1963–2021	Transformer	LSTM, GBDT	R^2_{OOS}	N	CRSP
Leippold and Wang (2022)	2022	China Eq.	2000–2020	LightGBM	Linear	R^2_{OOS}	N	DataStream
Avramov et al. (2023)	2023	US Eq.	1965–2020	Constrained NN	Linear	R^2_{OOS}	N	CRSP
Lopez-Lira and Tang (2023)	2023	US Eq.	2021–2023	GPT-3.5/4 zero-shot	FinBERT	Sharpe	Y	News
Chen et al. (2024)	2024	US Eq.	1957–2016	Deep ML factor	PCA, FF	Sharpe	N	CRSP

Table 7: Representative studies: Risk Management.

Study	Year	Asset	Period	ML Method	Bench.	Metric	Code	Data
Bollerslev (1986)	1986	–	–	GARCH(1,1) (bench.)	–	MSE	–	–
Engle (2002)	2002	Multi	–	DCC-GARCH (bench.)	–	MSE	–	–
Meinshausen (2006)	2006	–	–	Quantile RF	Quant. regr.	QC	Y	–
Holopainen and Sarlin (2017)	2017	Banks	1970–2014	RF	Logit	AUC	N	BIS
Barboza et al. (2017)	2017	US Corp.	1985–2013	RF, GBDT	Logit	AUC	Y	Compustat
Taylor (2019)	2019	Multi	1990–2017	NN-CAViaR	CAViaR	VaR cov.	N	Public
Zhang et al. (2019)	2019	SPX	1990–2018	GARCH-LSTM	GARCH	MAE	N	Public
Bucci (2020)	2020	SPX	2000–2019	LSTM	GARCH, HAR	RMSE	N	Oxford-Man
Luong and Dokuchaev (2021)	2021	SPX	2000–2019	RF	HAR-RV	MSE	N	Oxford-Man
Bussmann et al. (2021)	2021	Credit	2005–2019	GBDT + SHAP	Logit	AUC	Y	Private
Cont and Xu (2022)	2022	Multi	2005–2020	Tail-GAN	Hist. sim.	VaR cov.	N	Various
Mittnik et al. (2022)	2022	Indices	2000–2021	LightGBM	HAR-RV	QLIKE	N	Public
Giudici and Polinesi (2022)	2022	Banks	2008–2020	GNN	CoVaR	MES	N	Private
Suss and Trebbi (2022)	2022	Credit	–	Alt. data + ML	Trad. PD	AUC, KS	N	Private
Delli Gatti and Grazzini (2022)	2022	Macro	–	RL + ABM	ABM only	Crisis det.	N	Sim.
Christensen et al. (2023)	2023	Multi	2005–2022	TFT	HAR	QLIKE	N	Public

Table 8: Representative studies: Portfolio Optimization.

Study	Year	Asset	Period	ML Method	Bench.	Metric	Code	Data
Markowitz (1952)	1952	–	–	MV optimization (bench.)	–	Sharpe	–	–
Almgren and Chriss (2001)	2001	–	–	Analytical TC	–	Imp. short.	–	–
Jiang et al. (2017)	2017	Crypto	2015–2017	CNN-RL	UBAH	Cum. return	Y	Poloniex
Ban et al. (2018)	2018	US Eq.	1980–2014	ML + LASSO	MV	Sharpe	N	CRSP
Kelly et al. (2019)	2019	US Eq.	1962–2014	IPCA	FF5	Sharpe	N	CRSP
DeMiguel et al. (2020)	2020	US Eq.	1980–2014	LASSO + TC	EW	Sharpe (net)	N	CRSP
Zhang et al. (2020a)	2020	US Eq.	1990–2018	End-to-end NN	MV, EW	Sharpe	N	CRSP
Ye et al. (2020)	2020	US Eq.	2000–2019	PPO RL	EW, MV	Sharpe	N	CRSP
Nystrup et al. (2020)	2020	Multi	1990–2018	Regime-switch ML	MV	Sharpe	N	Public
Cong et al. (2021)	2021	US Eq.	1963–2019	Panel Tree	FF5	Sharpe	N	CRSP
Wang et al. (2021)	2021	US Eq.	2009–2021	FinRL framework	EW, Min-Var	Sharpe	Y	Yahoo
Ledoit and Wolf (2022)	2022	US Eq.	1972–2021	Nonlin. shrink. NN	Lin. shrink.	Sharpe	N	CRSP
Zhao et al. (2022)	2022	US Eq.	–	Graph cov.	Sample cov.	Sharpe	N	CRSP
Butler and Kwon (2023)	2023	US Eq.	–	Diff. CVX layer	MV	Sharpe	N	CRSP
Moallemi and Saglam (2023)	2023	US Eq.	–	DP + TC	Naive rebal.	TC-adj. Sharpe	N	–

Table 9: Representative studies: Derivatives Pricing and Hedging.

Study	Year	Asset	Period	ML Method	Bench.	Metric	Code	Data
Hutchinson et al. (1994)	1994	SPX opt.	1987–1991	Shallow NN	Black-Scholes	Pricing err.	N	–
Gatheral and Jacquier (2014)	2014	SPX opt.	–	SVI param.	–	Arb-free fit	N	–
Bayer and Stemper (2019)	2019	Synthetic	–	Deep NN	Monte Carlo	Calib. err.	N	N/A
Buehler et al. (2019)	2019	Synthetic	–	Deep hedging NN	Delta-gamma	CVaR P&L	Y	N/A
Becker et al. (2019)	2019	Synthetic	–	Deep optimal stop	Longstaff-Schwartz	Stop value err.	Y	N/A
Ackerer and Filipović (2020)	2020	SPX/SX5E opt.	–	Smoothing NN	SVI	Arb-free MSE	N	–
Ruf and Wang (2020)	2020	SPX opt.	–	Various NN	BS, Heston	Hedging err.	Y	–
Cao et al. (2021)	2021	Synthetic	–	DRL hedging	Delta-gamma	CVaR P&L	N	N/A
Horvath et al. (2021)	2021	Synthetic	–	Deep NN	PDE solver	Calib. err.	N	N/A
Murray et al. (2022)	2022	Synthetic	–	Deep NN multi-instr.	Delta-gamma	CVaR P&L	N	N/A
Cohen et al. (2023)	2023	SPX opt.	–	VAE	Heston, SVI	Surface RMSE	N	–
Gnoatto et al. (2023)	2023	XVA portf.	–	NN regression	Nested MC	XVA err.	N	–

Table 10: Representative studies: Market Microstructure.

Study	Year	Asset	Period	ML Method	Bench.	Metric	Code	Data
Almgren and Chriss (2001)	2001	–	–	Analytical exec.	–	Imp. short.	–	–
Guéant (2017)	2017	–	–	Avellaneda-Stoikov	–	PnL/inv.	–	–
Ntakaris et al. (2018)	2018	FI-2010	2010	SVM, ML baselines	–	Bench. ds.	Y	FI-2010

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Table 10 continued

Study	Year	Asset	Period	ML Method	Bench.	Metric	Code	Data
Spooner et al. (2018)	2018	LOB sim.	–	DRL (DQN+)	A-Stoikov	PnL/inv.	N	Sim.
Sirignano and Cont (2019)	2019	US LOB	2014–2018	DL universal	Per-stock DL	Pred. acc.	N	Private
Zhang et al. (2020b)	2020	LSE LOB	2007–2008	DeepLOB CNN/LSTM	Lin., SVM	Pred. acc.	Y	FI-2010
Cont and Kalinin (2021)	2021	US LOB	–	Hawkes/NPP	Stat. point proc.	LL	N	Private
Ning et al. (2021)	2021	US LOB	–	DDQN exec.	Almgren-Chriss	Imp. short.	N	Private
Fang et al. (2021)	2021	LOB sim.	–	Multi-agent RL	Single-agent	Imp. short.	N	Sim.
Gašperov and Kostanjcar (2021)	2021	LOB sim.	–	Multi-agent RL MM	Single-agent	PnL/inv.	N	Sim.
Schnaubelt (2022)	2022	US LOB	–	Meta-learning exec.	Static RL	Imp. short.	N	Private
Shi et al. (2022)	2022	LOB sim.	–	Normalizing flows	Hist. sim.	KL div.	N	–
Kolm et al. (2023)	2023	FI-2010	2010	Transformer	LSTM, CNN	Pred. acc.	Y	FI-2010

Table 11: Representative studies: Alternative Data and NLP.

Study	Year	Asset	Period	ML Method	Bench.	Metric	Code	Data
Bollen et al. (2011)	2011	DJIA	2008	Mood class.	Random	DA accuracy	N	Twitter
Loughran and McDonald (2011)	2011	US 10-Ks	1994–2008	LM dictionary	Harvard dict.	Class. acc.	–	10-Ks
Oliveira et al. (2017)	2017	US Eq.	2013–2016	SVM, RF	Naive	Class. acc.	N	StockTwits
Araci (2019)	2019	US news	–	FinBERT	BERT, dict.	F1	Y	Reuters

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Table 11 continued

Study	Year	Asset	Period	ML Method	Bench.	Metric	Code	Data
Qin and Yang (2019)	2019	Earn. calls	–	Multi-modal NN	Text only	Vol. pred.	N	Conf. calls
Ein-Dor et al. (2020)	2020	Fin. news	–	Wiki weak supv.	Supervised	F1	N	News
Renault (2020)	2020	Tweets	2009–2019	Transformer / dict.	BoW	ROC-AUC	N	StockTwits
Bradley et al. (2021)	2021	US Eq.	2018–2021	Sentiment cls.	–	Pred. ret.	N	Reddit
Katona et al. (2021)	2021	US retail	2011–2018	CNN sat. imagery	Analyst exp.	Sales pred.	N	Sat. data
Li et al. (2021)	2021	US Eq.	2001–2018	ML (Word2Vec+)	Dict.	Cult. score	N	SEC, calls
Mariko et al. (2022)	2022	Fin. news	–	Causal IE NN	Pattern match.	F1	N	FinCausal
Kalesnik and Luo (2022)	2022	ESG factors	–	ESG NLP scores	–	Sharpe	N	Mixed
Wu et al. (2023)	2023	Bloomberg	–	Bloomberg-GPT (50B)	GPT, BLOOM	FinNLP bench	N	Private
Yang et al. (2023)	2023	SEC, news	–	FinGPT (open)	FinBERT	F1	Y	Public
Lopez-Lira and Tang (2023)	2023	News	2021–2023	GPT-3.5/4 0-shot	FinBERT	Sharpe	Y	Headlines
Green et al. (2019)	2019	US Eq.	2008–2017	Glassdoor txt.	–	Ret. pred.	N	Glassdoor

Note on selection. The tables above include between 12 and 19 studies per domain, prioritized for: (i) methodological coverage (each row reflects a distinct method-benchmark pair), (ii) influence (citation-weighted), and (iii) availability in the public-data subset of the survey to facilitate replication. The choice of representative studies necessarily omits high-quality work; the full 227-row database in the GitHub repository is the authoritative version, and we welcome corrections via pull request.